THE BROADER GOAL: EVIDENCE SUFFICIENCY VS. UNCERTAINTY

A Decision Framework for Reduced-Lead ECG AI: Distinguishing Grounded Claims from Epistemic Abstention

SPOKEN SCRIPT: "***Our broader goal is to help researchers understand when reduced-lead ECG models have sufficient evidence to support their predictions —and when they should express uncertainty instead.***"

REDUCED-LEAD ECG TELEMETRY INPUT

Ambulatory Wearable or Emergency Transport (e.g., 6-Lead, 4-Lead, or 2-Lead Setup)

CORE QUESTION: Does the active sensor mask provide sufficient observability for the diagnosis?

REGIME A: SUFFICIENT EVIDENCE OBSERVED

ELECTROPHYSIOLOGICAL CONDITIONS MET:

- Primary anatomical leads required for pathology are ACTIVE.
- Example: Lead V1 retained in 4-lead setup capturing anterior septal vector.
- Cross-attention trace references physically verified sensors.

DESIRED CLINICAL AI BEHAVIOR

SUPPORT PREDICTION (HIGH CONFIDENCE)

CLINICAL SAFETY BENEFIT:

Clinician receives diagnostic recommendation with verified physical evidence.
Zero hallucination risk. Safe for emergency triage decisions.

REGIME B: PHYSICAL EVIDENCE ABSENT / DEGRADED

CRITICAL INFORMATION GAP IDENTIFIED:

- Vital anatomical projections are MISSING from the active leads.
- Example: Precordial chest leads V1-V6 missing in 2-lead limb telemetry.
- Model cannot physically verify the electrical vector of the pathology.

DESIRED CLINICAL AI BEHAVIOR

EXPRESS UNCERTAINTY & ABSTAIN

CLINICAL SAFETY BENEFIT:

Prevents false-positive diagnostic overconfidence. System flags:
'Insufficient Evidence in Current Leads - Acquire Standard 12-Lead ECG.'

RESEARCH IMPACT: Shifting Medical AI Evaluation from Raw Classification Accuracy to Observable Evidence Grounding